

plausibility diagnosis instead of simple presence checks. The STS4xA delivers a wide supply voltage range and has low current consumption, which improves the accuracy while consuming lower power. The STS41A-AW1B model is capable of supplying optional wettable flank packaging to support automatic optical inspection.

Sensirion

<https://sensirion.com>

Sound platform for hearables

Qualcomm's S5 sound platform is designed to provide premium audio in any environment. This Bluetooth Low



Energy audio module is based on a Bluetooth 5.3 dual-mode radio. The Snapdragon Sound technology delivers

premium audio in places where there could be interference from many other devices. This module uses third-generation Qualcomm adaptive active noise cancellation, which seamlessly adapts to wind noise and transparency depending on the users' environment and prevents unwanted noise. The device comprises a 32-bit programmable dual-core CPU with a clock speed of up to 80MHz, an external flash memory, and a dual-core 240MHz programmable DSP audio subsystem. The S5 platform is suitable for use in devices such as wireless earbuds, headsets, and speakers.

Qualcomm

<https://www.qualcomm.com/home>

ReDriver device for PCIe 4.0

The PI2EQX1624 from Diodes Inc. is a ReDriver device for PCI Express 4.0. It's the first device to run at 1.8V,



which allows it to have longer PCB traces. The lower operational voltage also enables it to work at reduced power consumption.

The device consumes about 27mW in the low-power mode while requiring around 288mW when at active mode. It provides programmable linear equalisation and flat

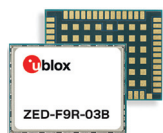
gain, by I2C control, to optimise performance over a variety of physical mediums by reducing interference. The PI2EQX1624 can be used in a wide field of applications, including data centres, servers, energy storage plants, and automobiles.

Diodes Inc.

<https://www.diodes.com>

High-precision GNSS module

U-blox's new global navigation satellite system (GNSS) module ZED-F9R-03B extends the range of positioning augmentation services. The device adds support for SPARTN 2.0 and



QZSS CLAS correction services, extending the geographical market reach and increasing the scalability of

applications using the module. It has an integrated inertial measurement unit (IMU) for RTK positioning. An in-built algorithm fuses the IMU data with GNSS measurements, wheel ticks, correction service data, and a vehicle dynamics model to provide centimetre-level positioning accuracy. The module supports GNSS bands like B1I, B2I, E1B/C, E5b, L1C/A, L1OF, L2C, L2OF. The module is ideal for use in applications where rapid access to highly accurate positioning data is crucial even in challenging signal environments.

U-Blox

<https://www.u-blox.com/en>

Powerful, precise chip resistors

TFHP series of chip resistors by TT Electronics combines power and precision in a single resistor. The chip



resistors are built using aluminium nitride (AlN) ceramic substrate. TFHP series can meet demanding thermal

management targets and provide a power of 2W in 1206 and 6W in 2512 chip sizes. These high-power resistors offer 0.1% tolerance, 25ppm/°C TCR combined with reduced self-heating, which improves linearity. The smaller

size and high power density of the TFHP series allow developers to build a compact and more reliable product.

TT Electronics

<https://www.ttelectronics.com>

T&M INSTRUMENTS

VNA-spectrum analyser

Anritsu VectorStar is a single-sweep VNA-spectrum analyser capable of supporting 70kHz to 220GHz. The VectorStar can conduct single connection VNA to verify active and



passive devices during the design, troubleshooting, or characterisation

stages. The test instrument also allows for spectral domain measurements of harmonics, spurious, other distortion products, and general frequency content. The VNA-spectrum analyser solution is ideal for on-wafer measurements and is compatible with all baseband VectorStar models. The T&M instrument allows engineers to analyse the VNA-like and spectrum-analyser-like responses of a device under test.

Anritsu

www.anritsu.com

Signal and frequency analysers

R&S FSV3050 and R&S FSVA3050 from Rohde and Schwarz are the new signal and frequency analyser instruments with an extended frequency range.



The T&M instruments offer a frequency range from 2Hz up to

50GHz, which is ideal for 5G NR testing. Both models support a full FR2 frequency range of up to 52.6GHz, making them suitable for use in the aerospace and defence industry. The models have an additional option that allows signal analysis of up to 54GHz. The instruments offer 1GHz analysis bandwidth, which enables the users to perform highly demanding measurement applications, such as linearising power amplifiers, capturing short